BOWIE STATE UNIVERSITY
School of Arts and Sciences
Department of Computer Science
COSC 631 Database Information Systems II (3 Credits)

Syllabus – Spring 2022

INSTRUCTOR: Dr. Bo Yang

Office Location: Computer Science Building, Room 322
Email: byang@bowiestate.edu
Phone: 2-4459 (in campus)
301-860-4459 (off campus)
Office Hours: Thu 3:00PM-5:00PM
Other time by appointment

PREREQUISITE: COSC531

REQUIRED TEXT:

- *Fundamentals of Database Systems*, Elmasri/Navathe; 7th or Later Editions, Addison-Wesley, ISBN **978-0133970777**

RECOMMENDED READING LIST:

- *Database Systems: Design, Implementation, and Management*, Carlos Coronel; 13th or Later Editions, Thomson, ISBN 978-1337627900

CATALOG DESCRIPTION:

To study the further database techniques beyond which covered in the COSC531, and thus to acquaint the students with some relatively advanced issues. At the end of the course students should be able to: gain an awareness of the advanced issues in relational data models, learn about the query optimization technology and applications, familiarize with the XML, data-warehousing and data-mining techniques and other advanced topics, apply the knowledge acquired to solve simple problems.

IMPORTANT REMINDERS:

Students who have a disability and want accommodations should report immediately to **Disability Support Services** (DSS), located in Room 311 of the Center for Learning and Technology or call Mr. Michael S. Hughes, DSS Coordinator, at 301-860-3292.

EPE Statement: Please take your **English Proficiency Examination** as early as possible! After completing ENGL 101 and ENGL 102, students must take and successfully pass the Bowie State University English Proficiency Examination. Transfer students who completed their English composition requirements at another university should take the English Proficiency Examination during their first semester of enrollment at the University.

Dept. of Comp Science (Secretary): 301-860-3961
Dept. of Comp Science (Fax): 301-860-3979
Bowie State University (Main): 301-860-4000

In case of inclement weather (snow etc.) please call 301-860-4000 to find out if the university is open.

TENTATIVE TIME SCHEDULE:

Class Meetings	Topics
Week 1: 1/26	*Syllabus and Further Discussion on Dependency*
Week 2: 2/2	*Query Processing and Optimization, Project#1*
Week 3: 2/9	*Transaction Processing*
Week 4: 2/16	*Database Recovery*
Week 5: 2/23	*Object-Oriented Database*
Week 6: 3/2	*Enhanced Data Models and Distributed Databases,* ***Quiz#1***
Week 7: 3/9	*Web Database,* ***Project#1 Due***
Week 8: 3/16	*Tentative Midterm*
Week 9: 3/23	*Spring Break*
Week10: 3/30	*XML Extensible Markup Language,* ***Quiz#2****, Project#2*
Week11: 4/6	*Data Mining Concepts*
Week12: 4/13	*Data Warehousing and OLAP*
Week13: 4/20	*Emerging Database Technologies*
Week14: 4/27	*Presentation & Demo*
Week15: 5/4	*Presentation & Demo,* ***Project#2 Due***
Week16: 5/11	*Final Exam Review,* ***Survey Paper Due***
Week17: 5/18	*Tentative Date of Non-Graduating Students* ***Final Exam***

TEACHING MODE:
Traditional lecture/demonstration and class discussion/project.

Note there will be quizzes given in the classes, and there is no make-up for past quizzes. Late homework, project or other assignments will have late penalty of 20% PER DAY. Make-up midterm or final exam will ONLY be allowed with prior approval of Professor. All assignments MUST be submitted to Blackboard instead of email.

EVALUATION METHOD:
Final semester grade will be determined by the average of the student performance on the homework, in-class projects, projects, midterm, and the final exam as follows:

- Class Participation 10%
- In-class Quiz & Homework 15%
- Paper 5%
- Two Projects 20%
- Midterm 20%

• Final	30%

Letter Grades will be assigned according to:

90 - 100	= A
80 - 89	= B
70 - 79	= C
60 - 69	= D
below 60	= F

IMPORTANT!

- Class attendance is mandatory at Bowie State University. In every class meeting you will be required to sign up. This policy will be enforced. After six (6) absences from a daytime course or two absences (2) from an evening course, the matter will be referred to the department chair. It is of importance that you attend **every** class!
- Expect a quiz or test every week. The quiz time will run from the start of class for 20 minutes. Therefore, if your are 20 minutes or more late, you will miss the quiz.
- All homework and assignments are to be turned in *on or before the due date*, even if class is cancelled for *any* reason or you are not able to attend class. Assignments may be submitted in class, slipped under my office door, or given to me during my office hours. It is each individual student's responsibility to submit homework and assignments on time.
- There will be **no make-up** for missed homework, assignments, or quizzes. Makeup of exams will be allowed *only* if the absence is validated by a formal documents (e.g., from the hospital) and a prior notice if it can be notified before the class.

Policy on Cell Phones, etc.

The ringers of cell phones, pagers and any other electronic devices must be turned OFF or set to vibrate during class time. Only calls of an urgent or emergency nature should be taken and you should step outside the classroom to do so. If you do not observe this policy, you may be asked to leave the class for that day and an unexcused absence will be incurred.